

Case Study of the Air India Express Flight 1344 - Air Crash Investigation

- **Flight:** Air India Express IX-1344
- **Aircraft:** Boeing 737-800 [Tail number: V T - AXH]
- **Date:** 7 August 2020 [covid]
- **Route:** Dubai → Kozhikode (Calicut), Kerala, India
- **Passengers:** 184
- **Pilot & Cabin Crew:** 6 (4 cabin crew & 2 pilot).
- **Fatalities:** 21 (including both pilots)
- **Injuries:** More than 100 injured
- **Location:** Kozhikode International Airport, Kerala
- **Pilot-in-Command:** Captain Deepak Vasant Sathe (59), an ex-Air Force pilot with **10,848 hours** of flying experience *Medical Note:* He was managing Type-2 diabetes and secretly taking unauthorized Ayurvedic medication which could cause low blood sugar, potentially clouding judgment
- **First Officer:** Akhilesh Kumar (32), with roughly **2,000 hours** of flying experience.

- **Brief about the Aircraft:**

The aircraft attempted to land during heavy monsoon rain. The first approach was abandoned. On the second approach, the aircraft landed far beyond the touchdown zone on the wet tabletop runway. Due to excessive speed, tailwind, and reduced braking effectiveness, the aircraft overran the runway, plunged down a slope, and broke apart.

- **Infrastructure Vulnerabilities:**

Kozhikode is designated as a **Category C airport** due to its risky **tabletop runway** (Runway 10/28) built on a flat hill with steep gorges at both ends

- **Runway Length:** Only **8,858 feet** long.
- **Poor Maintenance:** Years of neglect left rubber tire deposits clogging the runway surface, increasing water pooling during heavy monsoon rains
- **Safety Deviations:** The Runway End Safety Area (RESA) was only **240 feet** instead of the recommended **780 feet**

- **Flawed Instrument/Lighting:** Runway 10's approach lights extended only 150 meters (instead of 900 meters) Additionally, the runway's anemometer (wind measuring tool) was improperly calibrated and frequently faulty

- **Aircraft Details:**

Air India Express Flight 1344 was a Boeing 737-800, registration VT-AXH, manufactured by Boeing. It is a twin-engine, narrow-body commercial airliner belonging to the Boeing 737 Next Generation (NG) family. The aircraft was powered by two CFM56-7B turbofan engines. It was equipped with modern avionics, autopilot systems, weather radar, Ground Proximity Warning System (GPWS), and Traffic Collision Avoidance System (TCAS) to enhance flight safety.

- **Swiss Cheese Model Layers:**

- **Layer 1: Slice 1: Organizational & Administrative Pressures**

This included a packed schedule of the captain's rest day short to add another flight to his schedule the next day. This placed hidden pressure on him to avoid any substantial delays or diversions that would disrupt his upcoming roster. The airline's flight simulator training had known defects, and pilots were not being thoroughly trained to handle high-stress manual performance calculations under pressure.

- **Layer 2: Workplace & Environmental Conditions (Airport Infrastructure)**

The Runway End Safety Area (RESA) at Kozhikode was only 240 feet instead of the recommended 780 feet. Years of neglected rubber tire deposits clogged the surface, leading to severe water pooling and aquaplaning risks during monsoon rains. The runway's wind-measuring equipment (anemometer) was uncalibrated and frequently broken, causing the tower to give incorrect wind data to the pilots.

- **Layer 3: Unforeseen Operational Failures:**

Wiper Mechanical Failure: Just 27 seconds into the first landing approach, the captain's windshield wiper failed completely. This blinded his forward view during heavy downpours, leading to an aborted first landing.

Sudden Runway Change: The ATC changed the active runway from Runway 28 to Runway 10. This introduced a highly dangerous tailwind situation on a wet, compromised tabletop runway.

- **Layer 4: Tactical Decisions & Violations :**

Airline policy strictly prohibited landing in active rain with a broken wiper. The captain bypassed this safety barrier by hiding the failure from ATC and choosing to attempt a landing on Runway 10 instead of diverting to a clear alternate airport. Due to severe turbulence, the crew swapped their landing configuration from Flaps 40 to Flaps 30. While it gave better control, it critically increased the distance needed to bring the heavy aircraft to a complete stop.

- **Layer 5: Breakdown in Cockpit Resource Management:**

After disconnecting the autopilot, the captain left the auto-throttle active, overcorrected his altitude, and triggered proximity warning. The landing was no longer stable and required an immediate go-around. The communication gap between the pilot was an issue. The First Officer recognized the danger, calling out Captain and explicitly warning him. However, due to steep authority gradients or hesitation, he did not physically take control of the aircraft, and the captain ignored the warnings.

- **Post Reflections and actions taken:**

The aviation safety standards later changed on the:

- Strict tailwind compliance
- Enforcement of the wiper malfunction.
- Unstable approach discipline
- Strengthened cockpit teamwork.
- Simulator training calibration.
- Anemometer maintenance
- Runway Safety Area Oversight.

Video reference:

- ▶ **Air India Express Flight 1344 – Kozhikode Kerala Plane Crash Recreation | Real ...**
- ▶ **Tragedy of Air India Flight 1344: What REALLY Caused it?**